GNSS Integrity Lab

Interpretation of the Monte Carlo and real-data results

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1. Summary

The results show three things.

- **The RAIM implementation behaves as theory predicts.** False-alert and detection rates match the central and non-central chi-square models to within sampling error, and a bias placed at the protection-level value goes undetected at the designed missed-detection rate.
- **On an hour of real data, the monitor does its job.** It raises no false alarms, keeps errors well inside its protection levels, and detects and excludes a ramping single-satellite fault without ever outputting a position whose error exceeds the protection level.
- **The scenarios map the limits of the approach.** A common-mode clock push is invisible to RAIM by construction. The clock monitor catches fast pushes but not slow ones, and its alarm does not persist. A consistent position pull-off defeats both monitors and needs an independent reference to detect.

Table 1 summarises each test. Section 6 lists points to verify before presenting the results externally; the most important are what the position errors are measured against, and whether the error model was calibrated on the same hour it was tested on.

Test	Result	Verdict
False-alert rate (Monte Carlo)	9.92×10^{-4} vs 1×10^{-3} design	Matches design
Detection probability vs bias	Largest deviation from theory 0.003	Matches theory
Missed detection at the protection-level bias	1.01×10^{-3} vs 1×10^{-3} design	Matches design
Nominal real data	0 alarms; errors well below protection levels	As expected
A: single-satellite ramp	Detected at 12 m, excluded, no misleading output	Protected
B: common-mode time push	RAIM blind; clock monitor detects ≥ 0.1 m/s only	Partly protected
C: consistent position pull-off	Both monitors blind; 59 misleading epochs	Outside the fault model

Table 1. Findings at a glance.

2. Setup

- **Real data:** GSI (Japan) station 0759, 2 April 2005, one hour at 30 s intervals (120 epochs), from GPS RINEX observation and navigation files.
- **Positioning:** weighted least squares with an elevation-dependent error model, $\sigma^2 = 0.44^2 + (0.41 / \sin \text{el})^2$ m^2, inflated by 1.25 as an overbound.
- **RAIM:** chi-square test on the least-squares residuals with fault detection and exclusion, and slope-based horizontal and vertical protection levels (HPL, VPL). False-alert probability 1×10^{-5} per epoch on real data.
- **Clock monitor:** a Kalman filter on the receiver clock estimate, with drift and drift-rate process noise, tested through the normalised innovation squared (NIS) per epoch and over

a window. Tuned on the first 30 minutes and checked on the untouched second 30 minutes.

- **Monte Carlo:** 7 satellites (3 degrees of freedom), with false-alert and missed-detection probabilities both 1×10^{-3}.

The Monte Carlo and real-data runs use different false-alert settings (1×10^{-3} and 1×10^{-5}), so their detection thresholds differ. Both are illustrative values, not the allocations used in certified airborne receivers.

3. Monte Carlo validation

3.1 False alerts

Across 400,000 fault-free trials, the empirical false-alert rate was 9.92×10^{-4}, with a 95% confidence interval of 8.95×10^{-4} to 1.09×10^{-3}. The design value lies inside that interval. This confirms two things at once: the threshold is computed correctly from the central chi-square distribution with 3 degrees of freedom, and the residuals are normalised by the right sigmas. An error in either would shift the false-alert rate away from the design value.

3.2 Detection probability versus bias

Figure 1 compares simulated detection probability with non-central chi-square theory for two satellites: G11, which has the largest slope, and G08, which has the smallest. The simulated points lie on the theory curves across the whole range.

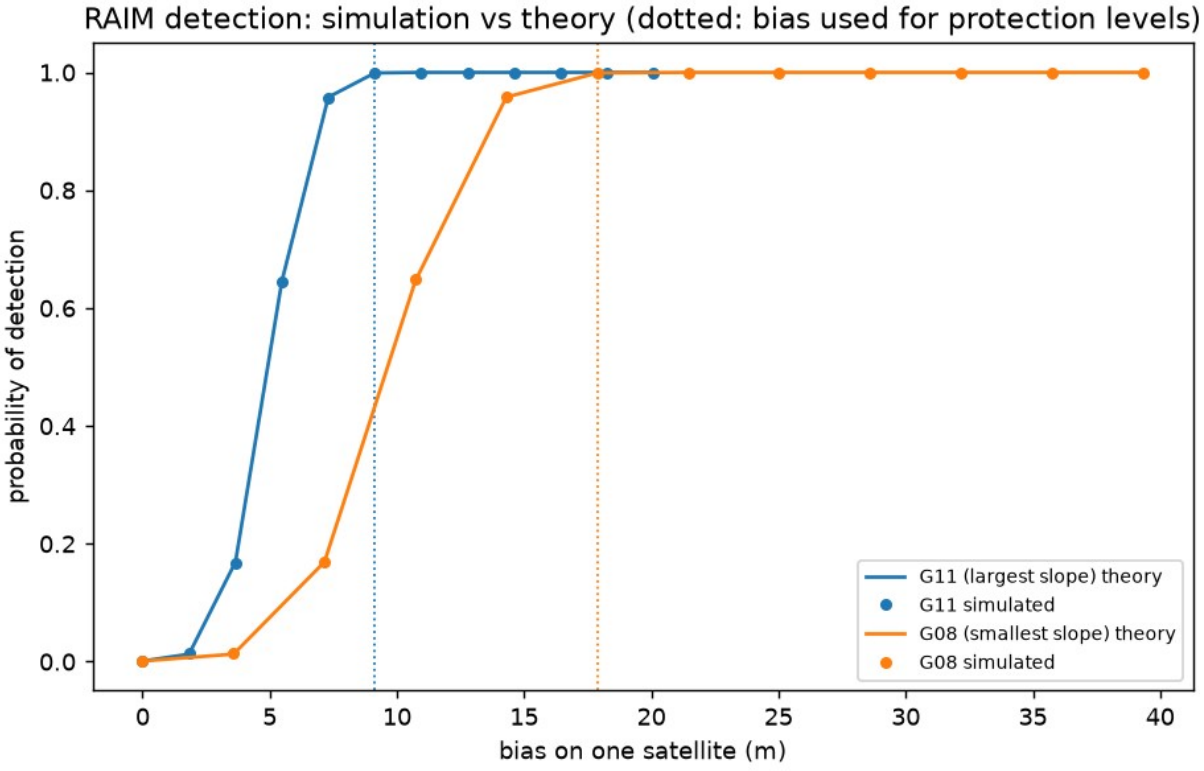


Figure 1. Probability of detection versus bias on one satellite: simulation (dots) against non-central chi-square theory (lines). Dotted lines mark the bias used for the protection levels.

The largest deviation from theory was 0.003. At a detection probability near 0.65 with 20,000 trials, one standard error is about 0.0034, so the agreement is within one standard error. (The summary file describes about 0.01 as sampling noise; the 95% band is closer to ±0.007, which the results still meet comfortably.)

The biases in the table are matched in difficulty: 5.5 m on G11 and 10.7 m on G08 both give a detection probability of about 0.65, so G08 needs about twice the bias to be detected. Detectability alone does not decide the risk, though. A satellite's slope is the horizontal error its fault causes per unit of signal in the test statistic. G11 is detected at smaller biases, yet it has the largest slope, so each metre of G11 bias moves the position by even more. G11 is therefore the satellite whose fault can do the most damage before being detected, and it sets the protection level.

3.3 Protection level

A bias of 9.1 m on G11 shifts the horizontal position by 7.9 m, which equals the HPL at these settings. That bias went undetected in 1.01×10^{-3} of trials (95% interval 9.09×10^{-4} to 1.11×10^{-3}), matching the missed-detection allocation. The protection level is therefore placed exactly where the allocation says it should be.

This HPL bounds only the shift caused by the fault. In least squares, the position noise is statistically independent of the residuals, so in the undetected cases at that bias the noise still adds to the shift: roughly half or more of them end with a total horizontal error above the HPL. This is why many formulations add a fault-free noise term to the protection level. Doing the same here, and re-running this check on total error, would close the gap.

4. Real-data results

4.1 Nominal performance

- No RAIM alarms in 120 epochs. At 1×10^{-5} per epoch, the expected number of false alarms is about 0.001, so zero is expected, but one hour cannot confirm a rate that low; that is the Monte Carlo's job.
- The test statistic peaked at 0.19 of the threshold, so it stayed at least five times below it throughout. That margin is consistent with the deliberately overbounded sigmas.
- Horizontal error averaged 0.40 m (maximum 0.91 m) against HPLs of 8.9–24.2 m, a margin of roughly ten or more. Vertical error averaged 0.53 m (maximum 2.23 m) against VPLs of 13.6–65.3 m. The larger VPLs reflect periods of weaker vertical geometry.
- In Figure 2A, the nominal statistic drifts upward from about 0.05 to 0.15–0.2 of the threshold in the second half hour. It remains far from the threshold, but it is worth relating to changes in the satellites in view.

4.2 Clock monitor calibration

The clock-estimate noise measured from second differences was 0.41 m, while the least-squares covariance implies 1.37 m. The output attributes the gap to the overbounded RAIM sigmas, but overbounding explains only a factor of 1.25. Removing it still leaves about 1.10 m, around 2.7 times the measured value.

A more likely explanation is time correlation. Much of the pseudorange error, such as multipath and residual atmospheric delay, changes slowly over many 30-second epochs. Second differences cancel slowly varying errors, so 0.41 m measures only the fast, epoch-to-epoch component. That is the right noise level for a monitor that tests epoch-to-epoch changes, but the explanation in the output should be updated.

On the held-out second half, the mean NIS was 0.85, against 1.0 expected for a correctly modelled scalar innovation. The filter is therefore slightly conservative, which is the safe direction. Zero false alarms in 60 epochs is consistent with the design, but too few epochs to confirm a low false-alarm rate.

4.3 Scenario A: ramp on one satellite

A ramp of 0.2 m/s was added to G11 from 30 minutes. It was detected after 60 s, when the bias had reached 12.0 m; at 30 s the 6 m bias was still below the detection threshold. The test statistic then rose to more than a thousand times the threshold (Figure 2A).

Fault detection and exclusion worked as intended. Of 58 alarmed epochs, G11 was correctly excluded in 51, the wrong satellite was never excluded, and in 7 the monitor alarmed but could not identify the faulty satellite, so it output no position. Those 7 epochs cost availability, which is the safe outcome. The largest horizontal error of any position actually output after onset was 6.2 m, probably at the undetected epoch at 30.5 minutes. That is below the HPL, and no epoch had an undetected error above the HPL. Without monitoring, the error would have reached 357 m.

Two details are worth checking in the logs. One is which epochs were ambiguous: likely those just after detection, when the bias is small relative to the noise. The other is the step at about 53 minutes, where the test statistic in panel A jumps and the HPL in panel C drops, consistent with a change in the satellites in view.

4.4 Scenario B: common-mode time push

Shifting every pseudorange by the same amount is absorbed entirely by the receiver clock estimate. The residuals do not change, so RAIM cannot see it by construction: the change in the test statistic was at the 10^{-9} level, which is numerical round-off. The position is also unaffected. The damage is to time, which matters to users who rely on GNSS for timing.

Ramp (m/s)	Rate (ns per s)	Clock alarm	Offset at alarm or end of run
0.02	0.07	Not detected	About 35 m (≈120 ns) by the end
0.05	0.17	Not detected	About 89 m (≈300 ns) by the end
0.1	0.33	After 30 s	3.0 m (10 ns)
0.2	0.67	After 30 s	6.0 m (20 ns)
0.5	1.67	After 30 s	15.0 m (50 ns)
2.0	6.67	After 30 s	60.0 m (200 ns)

Table 2. Clock monitor against time pushes. Offsets at the end of the run are derived from the ramp rate and the 29.5 minutes after onset.

Pushes of 0.1 m/s or faster were caught at the first epoch after onset. Slower pushes were never detected: they change the clock by only 0.6–1.5 m between epochs, within the innovation noise, and the filter's drift state gradually absorbs the new rate. Over the half hour they accumulated roughly 120 and 300 ns of timing error.

Figure 2B shows a second limitation. Even when a push is detected, the NIS spikes for a single epoch and then returns to normal while the offset keeps growing, because the filter re-converges on the pushed clock. A practical monitor would latch the alarm, check the estimated drift against what the oscillator can physically do, and add a cumulative test (such as CUSUM) to catch slow pushes.

4.5 Scenario C: consistent position pull-off

Here every pseudorange is altered consistently with a false position moving east at 0.5 m/s, with the clock kept consistent. The residuals stay self-consistent, so RAIM raised no alarm, and the clock was unaffected, so the clock monitor raised none either. By the end the horizontal error was 885 m against an HPL of 11 m, with 59 epochs of misleading information (Figure 2C).

This is the expected result, not a defect. Snapshot RAIM assumes faults on one or a few satellites that are inconsistent with the rest; a coherent change across all satellites lies outside

that fault model. Detecting it requires independent information, such as an inertial cross-check, signal-level checks, or multiple antennas.

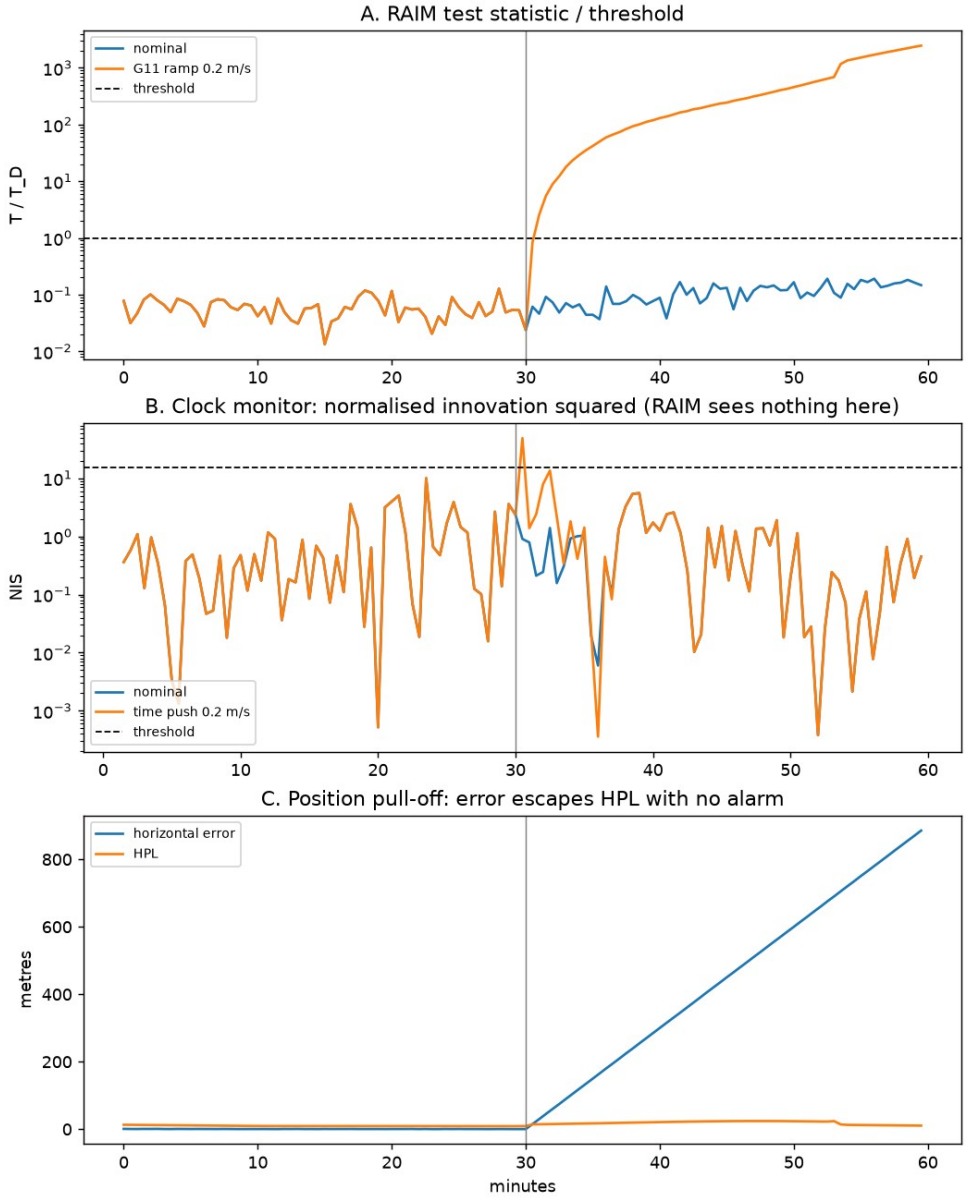


Figure 2. Real-data scenarios. A: RAIM test statistic relative to threshold, nominal and with a 0.2 m/s ramp on G11. B: clock monitor NIS, nominal and with a 0.2 m/s time push. C: horizontal error and HPL during a 0.5 m/s position pull-off.

5. What each monitor covers

Fault type	RAIM	Clock monitor	Outcome
Single-satellite ramp	Detects (here at 12 m) and excludes	Not designed for it	Protected
Common-mode time push, ≥ 0.1 m/s	Blind by construction	Detects at the first epoch, then goes quiet	Partly protected
Common-mode time push, ≤ 0.05 m/s	Blind by construction	Not detected	Timing unprotected
Consistent position pull-off	Blind	Blind	Unprotected; needs an independent reference

Table 3. Coverage of the two monitors across the tested fault types.

6. Caveats and points to verify

1. **Truth reference and observables.** State what the position errors are measured against and which pseudoranges and corrections are used. A sub-metre mean horizontal error is unusually good for single-frequency standalone GPS. If the reference comes from the same data, for example the session mean, the errors understate absolute accuracy.
2. **In-sample calibration.** The RAIM error model appears to be calibrated on the same hour it is tested on, which makes the no-alarm nominal result partly self-fulfilling. Calibrate on separate data, as was done for the clock monitor.
3. **Sample size.** One station and one hour cannot confirm rare-event rates or show that the overbound holds in general. Several days and stations would.
4. **Protection-level definition.** The HPL has no fault-free noise term (Section 3.3).
5. **Parameters.** The false-alert and missed-detection values are illustrative. Certified airborne receivers use allocations set by RTCA DO-229 and the applicable TSOs, so these results should not be presented as compliance with either.
6. **Clock monitor.** The alarm does not latch and slow pushes go undetected (Section 4.4), and overbounding does not explain the gap between the two clock-noise estimates (Section 4.2).
7. **Unexplained details.** The 7 ambiguous epochs, the step at about 53 minutes, and the upward drift of the nominal test statistic.

7. Suggested next steps

- Add a fault-free term to the HPL and VPL, and re-run the Monte Carlo check on total error rather than fault-induced shift.
- Calibrate the error model on separate data, then run several days and stations to measure false-alert behaviour and test the overbound.
- Log the satellite count, geometry and fault detection and exclusion decisions per epoch, to explain the ambiguous epochs and the 53-minute step.
- Latch clock-monitor alarms, add a drift plausibility check and a cumulative test for slow pushes, and plot detection time against ramp rate.
- Re-run with the false-alert and missed-detection allocations used for airborne receivers, to show how protection levels and availability change.
- For Scenario C, prototype a simple independent cross-check, such as a dead-reckoning or known-trajectory constraint, to show that the pull-off becomes detectable.